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CLOSED-LOOP ACTUATION, USER CONSENT CONTROLS, RESPONSE FEATURE EXTRACTION, AND SYSTEM CORRELATION USING HAPTIC-TAGGED WINDOWS (NON-LIMITING)

A. Overview (Non-Limiting)

In various embodiments, Node 1 supports optional vibration/haptic actuation in one or more modes including (i) guidance/notifications and (ii) user-enabled arousal activation/physiologic priming. This page describes non-limiting closed-loop control, consent gating, response feature extraction, and system correlation using actuation-tagged windows to improve interpretation reliability.

B. Consent Gating and User Control (Non-Limiting)

In various embodiments, vibration actuation—particularly activation/priming modes—is governed by explicit user controls. Non-limiting controls include: 1. explicit enablement within an application user interface or on-device gesture; 2. mode separation, wherein guidance haptics and activation haptics are independently enabled/disabled; 3. hard stop / abort control via gesture, app command, removal detection, or time-out; 4. bounded parameters (intensity limits, maximum continuous runtime, cumulative runtime per interval); and 5. context-aware lockouts, including automatic suppression during sleep-tagged windows unless explicitly enabled, and automatic disablement when integrity faults are present.

C. Actuation Protocol Definitions (Non-Limiting)

In various embodiments, Node 1 implements predefined or configurable actuation protocols. Non-limiting protocol families include: 1. baseline → actuation → recovery protocol with fixed phase durations; 2. ramp protocol that increases intensity or cadence to a ceiling then decreases; 3. burst protocol composed of pulsed bursts separated by rest intervals; 4. step protocol with discrete intensity steps; and 5. adaptive protocol that adjusts cadence/intensity based on quality indices and/or response features, within safety limits. In various embodiments, each protocol is represented internally as a sequence of time-stamped events and settings.

D. Closed-Loop Control and Adaptive Actuation (Optional, Non-Limiting)

In certain embodiments, Node 1 supports adaptive actuation that adjusts patterns in response to measured signals, while remaining safety-bounded. Non-limiting closed-loop inputs include: 1. PPG-derived features (pulse amplitude trends, pulse regularity/plausibility, derived indices); 2. pressure/contact stability (FSR/contact ring stability, contact integrity index); 3. strain trends (slow changes, stability, drift plausibility); 4. temperature context (to avoid actuation when temperature rises outside bounds); and 5. IMU stability (only actuate when motion stability is sufficient to reduce artifact and ensure comfort). In certain embodiments, the system uses a quality gate such that adaptive changes occur only when coupling and motion stability exceed thresholds.

E. Response Feature Extraction Using Actuation-Tagged Windows (Non-Limiting)

In various embodiments, the system computes response metrics using labeled windows including: 1. pre-actuation baseline window; 2. actuation window(s) (time-aligned to pulse patterns); and 3. post-actuation recovery window. Non-limiting response features include: 1. delta-from-baseline indices computed for selected signals; 2. response latency indices representing timing of detected changes after actuation onset; 3. response persistence indices representing whether changes sustain into recovery; 4. stability/confidence indicators derived from motion gating and coupling quality; and 5. artifact likelihood scores used to suppress or down-weight questionable responses. In certain embodiments, the device reports these outputs as abstract indices and confidence labels rather than raw waveforms.

F. Artifact Suppression Specific to Haptic Actuation (Non-Limiting)

In various embodiments, vibration can create mechanical artifacts in strain/pressure sensing and can disturb optical coupling in PPG. To mitigate this, Node 1 may: 1. blank sensitive channels during vibration pulses; 2. use interleaved sampling, where sensing occurs between pulses; 3. include a settling interval after actuation before resuming inference; 4. compute features from low-frequency components less affected by vibration; and/or 5. employ reference sensors (e.g., IMU vibration signature) to identify and subtract or gate contaminated samples.

G. Correlation with Other Nodes Using Actuation Markers (Non-Limiting)

In various embodiments, Node 1 actuation events are used as synchronized markers for correlation across nodes (including Nodes 2, 4, and 5), enabling system-level interpretation. Non-limiting correlation methods include: 1. using Node 1 actuation timestamps to align analysis windows across devices; 2. using Node 5 context tags (rest/active, activation state, thermal context) to condition interpretation of Node 1 response features; 3. comparing actuation response indices across repeated sessions under matched context; and 4. generating a system-level response confidence label based on multi-node agreement. In certain embodiments, the system uses Node 5 as an upper-body context anchor to ensure like-for-like comparisons of actuation response sessions.

H. Safety and Comfort Constraints for Activation Modes (Non-Limiting)

In various embodiments, activation/priming modes are constrained by: 1. maximum intensity, maximum duty cycle, and maximum duration; 2. thermal bounds and automatic disablement upon threshold exceedance; 3. contact integrity checks to avoid actuation under poor coupling; 4. motion-based suppression to prevent actuation during high movement; and 5. user-initiated start and immediate abort capability.

I. Variations

Any embodiment may modify protocol timing, actuator type, control logic, response features, and gating thresholds while maintaining the functional intent of consent-gated actuation, artifact-aware response measurement, and system correlation using actuation-tagged windows.